Reading panels of vision-language models share their errors: a reader study on radiology images

Anonymous

Abstract

Purpose: To measure error correlation among model readers and identify the sources of shared errors on radiology questions.

Materials and methods: We evaluated four application programming interface vision-language models (VLMs) and five open-weight models on five item sets from VQA-RAD, SLAKE, and MedXpertQA. We compared single readers, independent reading panels, and discussion panels. Manipulations included gray images, reduced resolution, region occlusion, image pairs, question-blind findings reports, and shared or private reference cases. Perceivers wrote reports; reasoners answered questions from them. Outcomes were error correlation and the effective number of independent readers. We calculated item bootstrap 95% confidence intervals.

Results: Panels of nine readers changed accuracy by −2.8 to 4.0 percentage points for GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5; Qwen 3.8 Flash gained 5.6 to 13.6 percentage points. Error correlation in generic panels was 0.442 to 0.915. Gray images reduced SLAKE single-reader accuracy by 25.2 to 40.0 percentage points; 64.2% to 76.2% of MedXpertQA-MM consensus errors repeated gray-image majorities. Gemini 3.7 Flash perceiver reports improved other reasoners' accuracy by 6.1 to 16.3 percentage points; shared-report error correlation was 0.800 to 0.990. On VQA-RAD, error correlation fell by 0.048 with roles versus 0.470 with different models. Discussion overturned 41.5% to 100.0% of correct minority answers.

Conclusions: A reading panel of nine readers built from one VLM provided 1.17 to 1.99 effective independent readers because its readers shared the model's seeing and the question-text prior. An independent second read of radiology images requires a different model to perform the seeing.

Keywords: Multi-agent system, Vision-language model, Double reading, Error correlation, Visual question answering, Visual grounding

1. Introduction

Double reading of screening mammograms increased cancer detection compared with a single reader.[1] In diagnostic radiology, a systematic review found that a second reader changed reports at rates that depended on the setting.[2] Radiologists' errors are perceptual or interpretive: a reader misses an image finding or assigns an incorrect meaning to a finding.[3] A second reader adds information only to the extent that the readers do not make the same errors. Pooling independent diagnostic judgments improved accuracy when readers had similar individual accuracy.[4] The value of another reader therefore depends on both individual accuracy and error correlation. Artificial intelligence (AI) systems have also been evaluated in the second-reader position, including a simulation of double reading in breast cancer screening.[5]

Multi-agent systems built from large language models emulate a clinical consultation. Published medical systems assigned clinical roles to several instances of a model, which answered independently before discussing their answers or voting.[6,7] A reading panel applies this arrangement to questions about radiology images. A large evaluation of agent systems found that the benefit of collaboration shrank as the underlying model became more capable.[8] Another evaluation found correlated errors across language models, including models with different architectures and providers.[9] In a reading panel built from one vision-language model (VLM), every reader shares the model's weights. These include the visual encoder that extracts image findings and the language model that selects the answer. Sampling and role prompts change the text of each reader's response while leaving the model and its image evidence shared. Different responses alone do not establish independent second reads.

VLMs answer many image questions using information in the question alone. The question-text prior is information supporting an answer from the question and options without image content. In general visual question answering, models exploited language priors; performance decreased when each question was paired with similar images requiring different answers.[10] This construction required the image to distinguish between answers to the same question. In medical visual question answering, leading multimodal models performed worse than random guessing on simple probing questions about diagnosis.[11] Analysis of medical image questions also found correct

answers accompanied by flawed image comprehension.[12] Replacing the image cost medical VLMs at most 6.5 percentage points of accuracy on medical benchmarks, against 48 to 61 percentage points on general-domain tasks,[13] and a radiology data set constructed to remove such shortcuts reduced text-only accuracy to near chance.[14] Answer accuracy alone does not measure the contribution of image evidence. In a reading panel, a prior that every reader shares is a common cause of the same error in all of them, and agreement among the readers does not show that each extracted the findings the answer requires.

Whether a reading panel of AI models provides independent second reads of radiology images has not been measured, nor have the sources of its shared errors. The aim of this study was to measure error correlation among AI readers answering radiology questions and determine its sources. We separated image evidence from the question-text prior and seeing from judging. Seeing was defined as the extraction of visual findings, and judging as the selection of an answer using findings and prior knowledge. We measured the contribution of each shared component: VLM, image rendering, role, reference cases, findings report, and discussion. The reader study evaluated four application programming interface (API) VLMs and five open-weight models on five item sets from VQA-RAD, SLAKE, and MedXpertQA.[15–17] Controlled manipulations varied the image through graded evidence, region occlusion, and paired images, and varied what readers shared, including the models used for seeing and judging (Fig. 1).

2. Material and methods

2.1. Study design

This reader study evaluated single readers and reading panels of AI VLMs answering closed questions about radiology images. Controlled manipulations varied the information available to readers and the components they shared. The design distinguished image evidence from the question-text prior and separated seeing from judging. The protocol and validation decisions were locked before the full runs. Pilot observations were excluded from the study analyses.

2.2. Data sets and sampling

Five item sets were assembled from public data sets (Table 1). S1 contained 250 of 251 eligible VQA-RAD test questions with yes/no answers.[15]

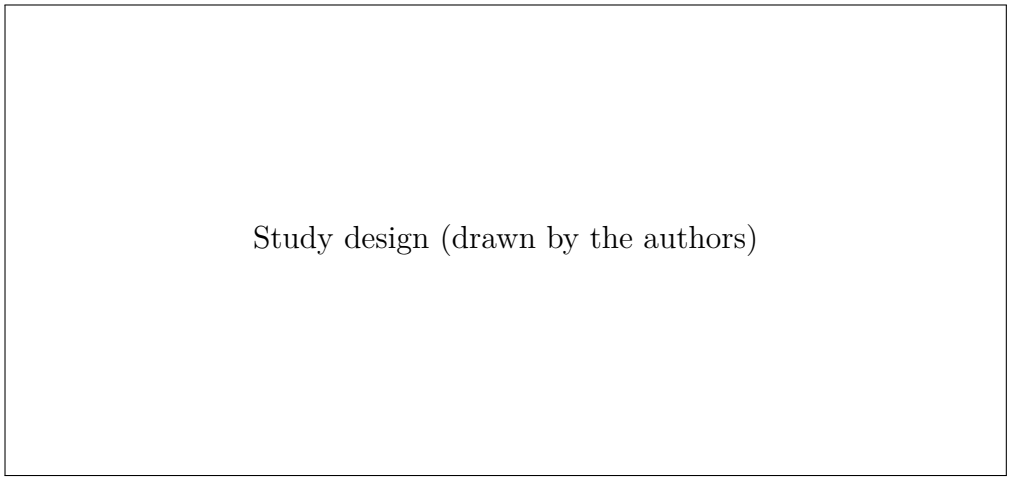


Figure 1: Study design across five item sets: VQA-RAD (S1), SLAKE (S2), MedXpertQA-MM (S3), SLAKE-target (S4), and SLAKE-pairs (S5). The comparisons include the single reader and reading panels: independent specialty panels, discussion panels, and generic panels. Image manipulations include graded evidence, region occlusion, and paired images; manipulations of what readers share include question-blind findings reports written by a perceiver and read by a reasoner, renderings, and reference cases. Outcomes are accuracy, error correlation, and the effective number of independent readers.

Duplicate image–question combinations were removed before sampling. S2 contained 250 of 355 eligible English SLAKE test questions labeled as closed and answered yes/no.[16] S3 contained 250 of 840 eligible MedXpertQA-MM items for which every attached image was a radiologic acquisition.[17] Image modality was labeled by GPT-4.1 nano (gpt-4.1-nano), and Claude Haiku 4.5 (claude-haiku-4.5) confirmed that candidate images were actual acquisitions. Drawings, diagrams, photographs, pathology slides, charts, and tables were excluded.

S4, SLAKE-target, contained 250 of 335 eligible English closed yes/no questions from the SLAKE training and test sets, excluding S2 items. Questions concerned an organ or lesion present in the image and localized by the supplied annotations. The expanded target bounding box occupied at most 40% of the image. Each item also had an equal-sized, nonoverlapping control box with at least 70% of its area inside the body. S5, SLAKE-pairs, contained 200 pairs, or 400 questions, spanning 104 question templates. Each pair used the same question with different images and opposite reference answers. Sampling allowed at most 5 pairs per template.

Sampling used stratified shuffling with seed 42. Strata were reference

answer for S1, reference answer and modality for S2, body system and medical task for S3, and target type and modality for S4. S5 was stratified by question template. Yes/no option order was randomized separately for each item using its identifier. Images were resized proportionally, without upsampling, to at most 1024 pixels on the long side.

Table 1: Item sets.

Set	Data set	Inclusion	Eligible	Sampled	Images	Options
S1	VQA-RAD	Test split, closed yes/no questions	251	250	135	Yes/no
S2	SLAKE	English test split, closed yes/no questions	355	250	93	Yes/no
S3	MedXpertQA-MM	Every attached image a radiologic acquisition; 48 items with 2–5 images	840	250	326	5
S4	SLAKE-target	English training and test splits, closed yes/no questions on an annotated organ or lesion; S2 excluded	335	250	196	Yes/no
S5	SLAKE-pairs	The same question on two images with opposite reference answers; 104 templates	–	200 pairs	252	Yes/no

2.3. Models

The API models were GPT-5 mini (gpt-5-mini), Gemini 3.7 Flash (gemini-3.7-flash), Claude Haiku 4.5, and Qwen 3.8 Flash (qwen3.8-flash). All reader calls used OpenRouter; the routing and modality-screening outputs of GPT-4.1 nano (temperature 0) were generated once and reused. Where a model accepted the parameter, panel readers, discussion rounds, and report writers sampled at temperature 0.7 and the single reader at 0.3; the seed was 1, and GPT-5 mini used low reasoning effort. Google models were restricted to Gemini 3.7 Flash. Routing and token budgets were held constant within each experimental condition. Because of its throughput, Qwen 3.8 Flash ran only the original-image baseline (A0), graded evidence (A1), generic reading panel (C1), and varied image rendering (C2).

Five local open-weight models were evaluated: MedGemma 4B, Qwen3-VL 8B, Qwen2-VL 7B, HuatuoGPT-Vision 7B, and LLaVA-1.5 7B.[18–22] HuatuoGPT-Vision 7B and LLaVA-1.5 7B shared the Contrastive Language–Image Pretraining (CLIP) ViT-L/14-336 vision encoder.[23] LLaVA-Med v1.5 was screened and excluded under the prespecified accuracy rule: its accuracy was 53%, below the 55% threshold, pooled over 100 validation items. These comprised 50 items each from S1 and S2. LLaVA-1.5 7B met the pooled threshold; its VQA-RAD validation accuracy was near chance.

Table 2: Experiments, conditions, readers, and models. CLAHE indicates contrast-limited adaptive histogram equalization.

Experiment	Sets	Conditions	Readers	Models
A0, original images	S1–S3	Original image	Single reader; specialty panels of 1, 3, 9; discussion panels of 3, 9	4 API
A1, graded evidence	S1–S3	256 and 128 pixels; uniform gray image	Single reader; specialty panels	4 API
A2, region occlusion	S4	Target box; control box	Single reader; specialty panels	3 API
A3, image pairs	S5	Original images	Single reader; specialty panels	3 API
B, seeing and judging	S1–S3	Question-blind findings reports by 3 perceivers; each reader's own report or one shared report	Panels of 9 report readers	3 perceivers $\times$ 3 reasoners
C, shared components	S1–S3	Generic panel; renderings (original, gamma 0.7, CLAHE); reference cases (S1–S2: shared or private, with images or as text)	Generic panels of 9; single reader with reference cases	4 API (reference cases: 3 API)
D, open-weight models	S1–S5	Original, gray, 256 and 128 pixels, gamma 0.7, CLAHE; target and control occlusion (S4)	One forward pass per item	5 open-weight

2.4. Readers and reading panels

The single reader received a general-radiologist role and instructions to reason briefly, step by step, before selecting an answer. For each specialty panel, GPT-4.1 nano ranked 9 radiology subspecialty roles using the question text. The roster covered neuroradiology, head and neck, thoracic, cardiovascular, abdominal, genitourinary, musculoskeletal, breast, pediatric, nuclear medicine, interventional, and emergency radiology. Panels of 1, 3, and 9 readers were nested: smaller panels used the first members of the same ranked roster and reused their initial answers. Answers were aggregated by majority vote.

Discussion panels started from these independent answers and allowed up to 3 subsequent rounds, stopping at unanimity. Readers reviewed peer answers and reasoning and could retain or revise their answers. Peer context was capped at 900 characters, with peer order shuffled by item and round. Generic panels contained 9 readers with identical general-radiologist prompts; their initial responses differed only through separate sampling.

2.5. Experiment A: image evidence and question-text prior

Experiment A used S1–S5. The original-image baseline compared the single reader, independent specialty panels, and discussion panels of 3 and 9 readers. Graded evidence compared original images capped at 1024 pixels, images capped at 256 or 128 pixels, and uniform gray images. Gray images retained the original rendered dimensions, and prompts and options remained unchanged. The single reader and independent specialty panels were evaluated at each evidence level.

On S4, the annotated target bounding box was expanded by 10% on each side and clipped to image boundaries. Target and control boxes were filled with mid-gray before resizing. Control boxes matched target-box dimensions, did not overlap the target, and met the body-coverage criterion. On S5, readers answered each member of the same-question, different-image pairs separately using the original images.

2.6. Experiment B: seeing and judging

Experiment B used S1–S3 and crossed 3 perceiver models with 3 reasoner models: GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5. Each perceiver received images without questions or answer options. It wrote findings reports of at most 120 words describing modality and plane, anatomy shown, and every abnormal or notable finding with its location. Reports were reused

across questions about the same image. Each perceiver wrote its 9 reports for each image once, before any reasoner read a report. Every reasoner therefore read identical reports for a given perceiver, image, and slot. In total, 3 perceivers wrote 9 reports for each of 478 images or, for multi-image items, image sets (12,906 reports).

Reasoners received the report, question, and options without the image. In the independent-report condition, each of 9 readers received its own report; all perceiver–reasoner combinations were evaluated. In the shared-report condition, all readers received the first report; this condition used matching perceiver and reasoner models. Generic panels reading images directly provided the reference comparison. For S2, report coverage of the queried organ or lesion was assessed using a synonym list fixed from SLAKE labels before running this pathway.

2.7. Experiment C: shared components

Experiment C used S1–S3 to vary shared VLMs, image rendering, roles, reports, reference cases, and discussion records. In the rendering condition, generic readers 1–3 viewed original images, readers 4–6 viewed gamma-transformed images with gamma 0.7, and readers 7–9 viewed CLAHE images.[24] CLAHE used a clip limit of 2.0 and an 8×8 tile grid on the lightness channel. Gamma 0.7 replaced grayscale inversion because inversion reduced mean single-reader accuracy by 6.6 percentage points in validation, exceeding the prespecified 5-percentage-point limit. Cross-vendor comparisons combined existing reader responses without additional calls.

Reference cases were evaluated on S1 and S2 using VQA-RAD and English SLAKE training images with closed yes/no questions. Images identical to any test image were excluded by SHA-256 hash. For each query, exclusions also removed reference cases within Hamming distance 8 of its 64-bit perceptual hash or with CLIP cosine similarity of at least 0.98. The data sets carried no patient identifiers; these bounds guarded against reference cases from the same scan or patient.

Retrieval used CLIP ViT-L/14-336 image embeddings. The 27 nearest eligible images were shuffled deterministically by question identifier into 9 sets of 3. For shared reference cases, set 1 was supplied to every reader; for private reference cases, set i was supplied to reader i. Each reference case included its image and a closed question–answer pair. The question was selected by maximal term frequency–inverse document frequency (TF-IDF) cosine similarity to the current question. Text-only reference cases supplied

the same question and answer without the reference image. Single readers also received image-based or text-only reference cases.

2.8. Experiment D: open-weight models

Experiment D used S1–S5 and the same rendered pixels as the API arm. Conditions included original and gray images, 256- and 128-pixel images, gamma 0.7, CLAHE, and target and control occlusion on S4. S5 used original images only. Each item and condition required one forward pass without sampling. Option probabilities were obtained by applying softmax to next-token logits for the answer words yes/no on binary items and option letters otherwise. The highest logit among accepted capitalization and leading-space token variants represented each answer. Letter readout for yes/no items was rejected during validation because the LLaVA family assigned 96%–100% probability to "A" regardless of image content.

2.9. Outcome measures

Accuracy was the proportion of answers matching the reference. Error correlation, ϕ, was the mean Pearson correlation of binary error indicators across items over reader pairs. Pairs with constant error indicators were omitted. The effective number of independent readers was $N_{\text{eff}} = 9/(1+8\phi)$. For readers of unequal accuracy, correlation was also divided by its maximum given their marginal error rates, $\phi_{\max}$.

The accuracy difference between a reading panel and the single reader was divided into two parts that summed to it. The first was the mean accuracy of the panel's readers minus the single reader's accuracy. The second was the panel's majority-vote accuracy minus the mean accuracy of its own readers (aggregation). Both differences were paired by item. For every reading panel of 9 readers on yes/no items, observed majority-vote accuracy was compared with accuracy predicted by a beta-binomial model. This comparison covered VQA-RAD, SLAKE, SLAKE-target and SLAKE-pairs across all conditions. The model's mean was the readers' mean accuracy, and its intraclass correlation was their error correlation. A majority of 9 required at least 5 correct readers.

The prior decomposition cross-tabulated each reader's correctness with real and gray images. Consensus errors required at least 7 of 9 readers to be wrong; their majority answers were compared with gray-image majorities. Flip rate measured the proportion of answers changed by occlusion. Region

dependence was target flip rate minus control flip rate, calculated for readers and panel majorities. Pair accuracy required both paired answers to be correct; the proportion of pairs answered differently was also measured. The two answers of each pair were compared as option text (yes or no) because option order was randomized separately for each item. Report-pathway analyses compared accuracy and error correlation and separated errors by report coverage of the queried target. Reader and panel accuracies from reports were compared with those of the generic panel of the same reasoner model reading images directly. Differences were paired by item.

The primary analysis compared matched groups of reader pairs within each reasoner model using differences in mean pairwise error correlation. The role contrast was the mean for reader pairs in specialty panels minus that for reader pairs in generic panels. The image rendering contrast was the mean for pairs that saw different renderings minus that for generic-panel pairs. The reference cases contrast was the mean for pairs with private reference cases minus that for pairs with shared reference cases. The report contrast was the mean for pairs that each read their own report from the same model minus that for pairs reading one shared report. The perceiver model contrast was the mean for pairs reading reports from two different perceivers minus that for pairs reading different reports from one perceiver. The reasoner model contrast compared pairs of two reasoner models reading the same report with pairs of one reasoner model reading one shared report. The model contrast compared generic-panel readers of two models with generic-panel readers of one model. Both contrasts subtracted the mean for pairs of one model, averaged over the two models. The baseline was the error correlation between generic-panel readers of one model, which shared the model, image, rendering, role, and reference cases. Each contrast was computed for each model or model pair and averaged over models. An ordinary least squares regression of pairwise error correlation on indicators of the components two readers did not share summarized the same comparisons. The regression included one intercept per pair of reasoner models. Discussion was evaluated separately by comparing round-0 and final error correlation. Conformity was the proportion of readers correct in round 0 who ended the discussion wrong. It was stratified by whether their round-0 answer differed from the same reader's answer with the gray image. It was also stratified by whether the reader's round-0 answer was in the round-0 majority.

For open-weight models, visual reliance was the absolute change in log-odds of yes between real and gray images. For multiple-option items, visual

reliance was the Kullback–Leibler divergence from the real-image to gray-image probability distribution. Region reliance subtracted the corresponding control-occlusion change from the target-occlusion change. Model-pair error correlations were compared by shared vision encoder, and Spearman correlation related API consensus errors to open-weight visual reliance.

2.10. Statistical analysis

Percentile bootstrap 95% confidence intervals (CIs) used 2,000 item-level resamples with seed 0, including paired differences. Error-correlation intervals used 500 resamples; the regression, the matched contrasts, and the comparison of round-0 with final error correlation used 300 resamples. Matched contrasts were computed within the same resampled item sets as the regression. Resampling retained all reader responses and matched conditions for each item; paired-image measures resampled complete pairs. Unparsed answers counted as wrong. Abstentions (answers that could not be parsed, mostly refusals to read the gray image) were reported for each condition, and error correlation was also computed on the items both readers answered, as a sensitivity analysis. Analyses used descriptive estimates and CIs without hypothesis tests or P values.

2.11. Validation before full runs

V1 checked API answer parsing and reasoning effort; V2 checked image transformations; V3 checked reference retrieval, leakage exclusions, and multiple-image parsing. V4 checked target and control boxes and image pairs; V5 checked report length, completeness, and answer parsing; V6 checked local-model probabilities and accuracy eligibility. Full runs used only the settings fixed by these checks.

2.12. Code and data availability

The code is public at `https://github.com/hq0709/consultation-arithmetic`. VQA-RAD, SLAKE, and MedXpertQA are public data sets.

3. Results

3.1. Reading panels and error correlation

Panels of nine readers changed accuracy little relative to the single reader for GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5. Reading-panel differences ranged from −2.8 to 0.8, −1.2 to 1.2, and −2.8 to 4.0 percentage

points, respectively (Fig. 2a). Among their reading and discussion panels, only the Claude Haiku 4.5 discussion panel on VQA-RAD had a difference CI excluding zero: 5.6 percentage points (95% CI, 0.8 to 10.8).

Accuracy gains were larger for Qwen 3.8 Flash reading panels than for the other models' reading panels. The VQA-RAD difference was 13.6 percentage points (95% CI, 7.6 to 19.6). Differences were 8.0 percentage points (95% CI, 4.0 to 12.4) on SLAKE and 5.6 percentage points (95% CI, 0.4 to 11.2) on MedXpertQA-MM. Discussion-panel differences ranged from 8.8 to 14.0 percentage points, and all their CIs excluded zero (Fig. 2a).

Higher accuracy among Qwen 3.8 Flash panel readers accounted for more of the panel advantage than majority voting did. Mean reader accuracy exceeded single-reader accuracy by 3.1 to 10.3 percentage points; majority voting added 1.1 to 3.3 percentage points relative to those readers. For the other models, majority voting changed accuracy by −2.1 to 2.4 percentage points relative to mean reader accuracy (Supplementary Tables).

Panels behaved like few independent readers. The effective number of independent readers ranged from 1.17 to 1.39 for GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5. It ranged from 1.46 to 1.99 for Qwen 3.8 Flash (Fig. 2b).

Discussion brought reader errors close to complete correlation, while unanimous answers still included errors. Final error correlations ranged from 0.995 to 1.000 for the first three models and from 0.822 to 0.913 for Qwen 3.8 Flash (Fig. 2c). Unanimity after discussion ranged from 82.4% to 100.0%, and 8.9% to 70.0% of unanimous answers were wrong.

The beta-binomial model closely reproduced majority accuracy on yes/no items. Across 108 panels, observed accuracy minus the prediction averaged −0.07 percentage points, with a standard deviation (SD) of 0.84 percentage points (Fig. 2d). The largest absolute difference was 2.59 percentage points. Majority accuracy minus mean reader accuracy ranged from −2.1 to 5.6 percentage points across error correlations of 0.44 to 0.99 (Fig. 2e).

3.2. Image evidence and the question-text prior

Replacing the image with gray lowered accuracy on VQA-RAD and SLAKE. Single-reader accuracy ranged from 60.8% to 81.2% with original VQA-RAD images and from 46.8% to 60.8% with gray images (Fig. 3a). Corresponding SLAKE ranges were 72.4% to 91.2% and 38.8% to 54.4%. Paired accuracy differences ranged from 11.2 to 26.8 percentage points on VQA-RAD and from 25.2 to 40.0 percentage points on SLAKE. All

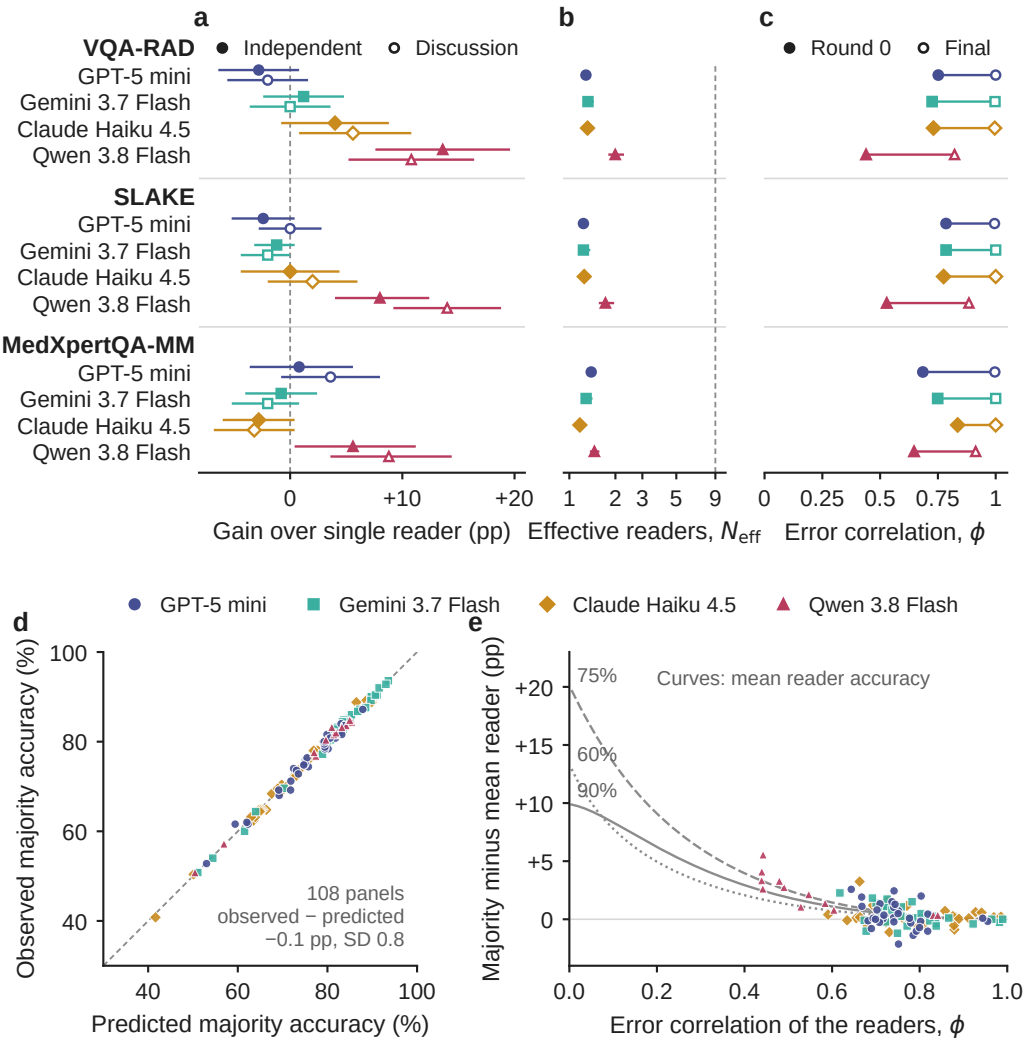


Figure 2: Reading-panel accuracy and error correlation. (a) Accuracy differences for independent specialty panels and discussion panels, each with 9 readers, relative to the single reader. (b) Effective number of independent readers, N_{eff}. (c) Round-0 and final error correlations, ϕ. Rows group VQA-RAD, SLAKE, and MedXpertQA-MM, with 250 items each; matched Qwen 3.8 Flash analyses use 250, 250, and 250 items, respectively. (d) Observed versus beta-binomial predicted majority accuracy for 108 panels on VQA-RAD, SLAKE, SLAKE-target (250 items each), and SLAKE-pairs (200 pairs). (e) Majority accuracy minus mean reader accuracy versus ϕ. Filled and open markers denote independent specialty and discussion panels in (a), and round-0 and final estimates in (c); lines join paired estimates in (c). Blue circles denote GPT-5 mini; teal squares, Gemini 3.7 Flash; amber diamonds, Claude Haiku 4.5; rose triangles, Qwen 3.8 Flash. Dotted, dashed, and solid curves in (e) represent predictions at the labeled mean accuracies. Dashed reference lines mark no accuracy difference in (a), 9 independent readers in (b), and equality in (d); the horizontal reference in (e) marks no aggregation gain. Whiskers in (a) are paired item bootstrap 95% CIs; those in (b) are transformed item bootstrap 95% CIs for ϕ. pp indicates percentage points.

single-reader and reading-panel difference CIs excluded zero on both data sets.

Removing the image changed single-reader accuracy little on MedXpertQA-MM for GPT-5 mini, Claude Haiku 4.5, and Qwen 3.8 Flash. Their paired differences ranged from 2.0 to 4.0 percentage points, and all CIs included zero. Gemini 3.7 Flash had a difference of 18.0 percentage points (95% CI, 12.0 to 23.6). Reading-panel difference CIs excluded zero for GPT-5 mini, Gemini 3.7 Flash, and Qwen 3.8 Flash (Supplementary Tables).

Removing the image increased error correlation for Gemini 3.7 Flash, Claude Haiku 4.5, and Qwen 3.8 Flash on VQA-RAD and SLAKE. Their VQA-RAD correlations ranged from 0.439 to 0.731 with original images and from 0.841 to 0.922 with gray images (Fig. 3b). Corresponding SLAKE ranges were 0.529 to 0.785 and 0.831 to 0.982. Error correlation did not change monotonically with image resolution (Fig. 3b).

Claude Haiku 4.5 readers remained highly correlated when gray-image analyses excluded abstentions. Abstentions ranged from 4.3% to 14.8% across data sets. Error correlation among answered items was 0.963 (95% CI, 0.942 to 0.982) on VQA-RAD and 0.931 (95% CI, 0.901 to 0.956) on SLAKE. It was 0.852 (95% CI, 0.803 to 0.891) on MedXpertQA-MM.

More reader answers were correct only with the real image than only with the gray image. On VQA-RAD, these shares ranged from 24.1% to 37.3% and from 10.4% to 13.6%, respectively (Fig. 4a). Corresponding SLAKE ranges were 36.3% to 46.4% and 4.4% to 15.8%. On MedXpertQA-MM, they were 8.1% to 22.3% and 4.4% to 7.9%; 12.3% to 64.0% of answers were wrong with both images.

Most consensus errors on MedXpertQA-MM repeated the gray-image majority answer. The share ranged from 64.2% to 76.2%, compared with 42.9% to 54.4% on VQA-RAD and 14.3% to 57.9% on SLAKE (Fig. 4b).

3.3. Use of the queried region and of the image

API readers changed answers more often after target occlusion than after control occlusion, with the largest difference for Gemini 3.7 Flash. On SLAKE-target, region dependence was 17.1 percentage points (95% CI, 13.2 to 21.2) for GPT-5 mini (Fig. 5a,b). It was 32.9 percentage points (95% CI, 28.2 to 38.0) for Gemini 3.7 Flash and 3.2 percentage points (95% CI, 0.8 to 5.7) for Claude Haiku 4.5.

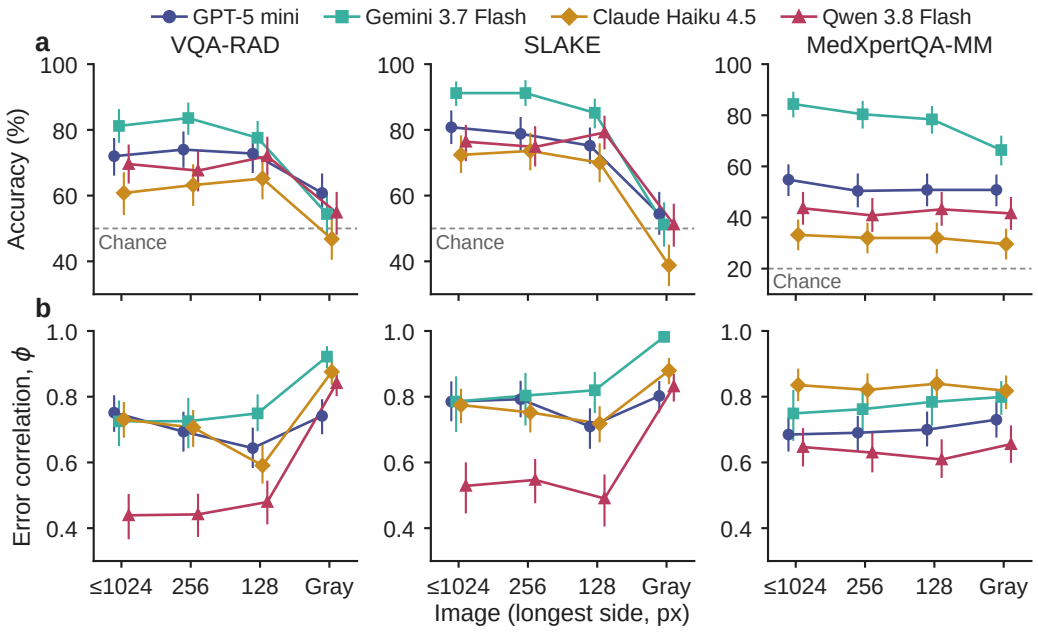


Figure 3: Accuracy and error correlation across image evidence levels. (a) Single-reader accuracy. (b) Error correlation, ϕ, among independent specialty-panel readers. Columns show VQA-RAD, SLAKE, and MedXpertQA-MM (250 items each). Image conditions are original images capped at 1024 pixels on the longest side, images capped at 256 or 128 pixels, and uniform gray images of unchanged dimensions. Blue circles denote GPT-5 mini; teal squares, Gemini 3.7 Flash; amber diamonds, Claude Haiku 4.5; rose triangles, Qwen 3.8 Flash. Lines join adjacent available conditions; missing estimates are omitted. Dashed horizontal lines in (a) mark chance accuracy: 50.0% for VQA-RAD and SLAKE and 20.0% for MedXpertQA-MM. Vertical intervals are item bootstrap 95% CIs. px indicates pixels.

Region dependence was positive across open-weight models, with the LLaVA-1.5 7B interval including zero. Its region dependence was 2.4 percentage points (95% CI, −2.0 to 6.8); all other open-weight model CIs excluded zero (Fig. 5b).

Pair accuracy was highest for Gemini 3.7 Flash among the API models tested. On SLAKE-pairs, both single-reader answers were correct for 59.0% (95% CI, 52.0 to 65.5) of pairs with GPT-5 mini (Fig. 5c). Rates were 74.0% (95% CI, 67.5 to 80.0) with Gemini 3.7 Flash and 38.5% (95% CI, 31.5 to 45.0) with Claude Haiku 4.5. Their answers differed between paired images in 62.0%, 75.0%, and 42.5% of pairs, respectively (Fig. 5c; Supplementary Tables).

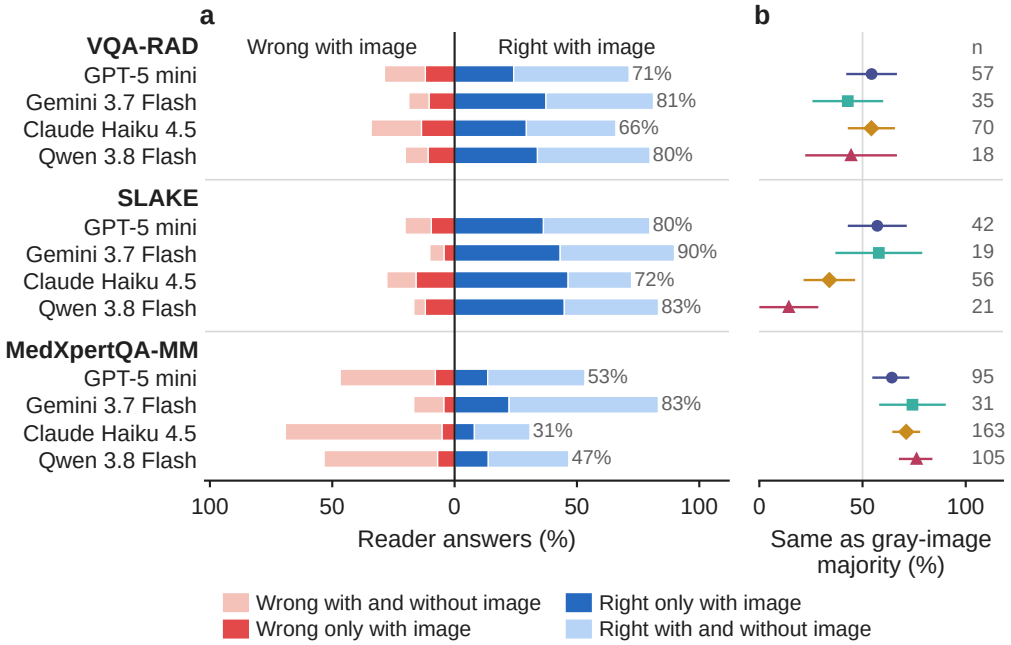


Figure 4: Reader correctness with real and gray images and agreement of consensus errors with gray-image majorities. (a) Reader answers from independent specialty panels of 9, cross-tabulated by correctness with each image. (b) Proportion of consensus errors matching the gray-image majority. Consensus errors require at least 7 of 9 readers to be wrong. Data sets are VQA-RAD, SLAKE, and MedXpertQA-MM (250 items each). In (a), dark blue denotes correct only with the real image; light blue, correct with both; dark red, wrong only with the real image; light red, wrong with both. End labels give real-image accuracy. In (b), blue circles denote GPT-5 mini; teal squares, Gemini 3.7 Flash; amber diamonds, Claude Haiku 4.5; rose triangles, Qwen 3.8 Flash. The vertical zero line in (a) separates real-image errors on the left from correct answers on the right; the vertical reference in (b) marks 50.0%. Horizontal intervals in (b) are item bootstrap 95% CIs. n indicates consensus-error counts.

3.4. Seeing and judging

Reader accuracy depended on both the perceiver and the reasoner when each reader read its own question-blind findings report. Across reasoners, accuracy with Gemini 3.7 Flash reports ranged from 78.7% to 81.3% on VQA-RAD and from 84.3% to 88.2% on SLAKE (Fig. 6a). Corresponding ranges with GPT-5 mini reports were 69.0% to 70.6% and 81.0% to 84.9%; with Claude Haiku 4.5 reports, 61.6% to 63.6% and 76.3% to 80.2%.

Gemini 3.7 Flash reports raised accuracy for the other reasoners relative

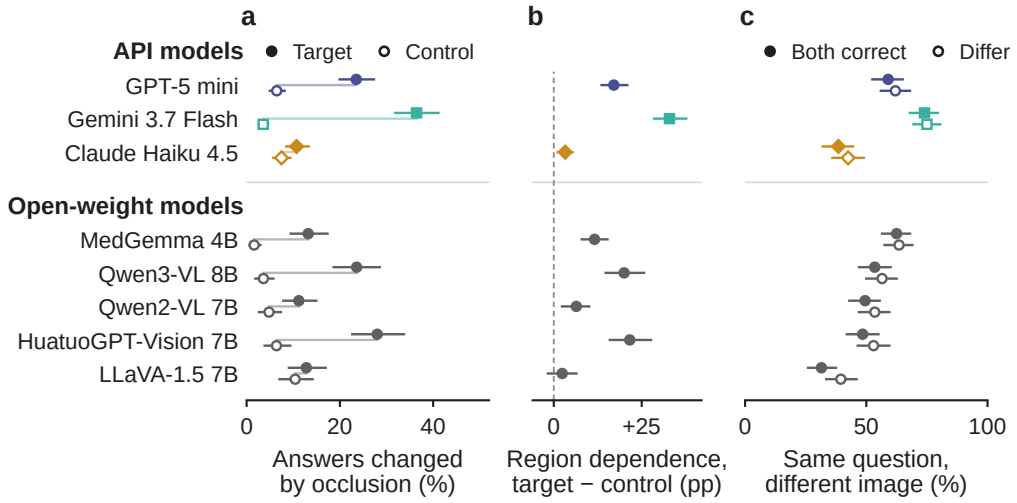


Figure 5: Answer changes after region occlusion and between paired images. (a) Target and control occlusion flip rates on SLAKE-target (250 items). (b) Region dependence, target minus control flip rate. (c) Pair accuracy and proportion of answers differing on SLAKE-pairs (200 pairs with identical questions and opposite reference answers). API occlusion estimates average panel-reader responses within items; paired-image estimates use single readers. Open-weight models use one forward pass per item. Filled and open markers denote target and control occlusion in (a), and both answers correct and answers differing in (c). Lines in (a) join the rates. Blue circles denote GPT-5 mini; teal squares, Gemini 3.7 Flash; amber diamonds, Claude Haiku 4.5; dark gray circles denote the labeled open-weight models. The dashed vertical line in (b) marks zero. Horizontal intervals are item bootstrap 95% CIs, with complete pairs resampled in (c). pp indicates percentage points.

to reading the image. Paired gains ranged from 6.1 percentage points (95% CI, 1.4 to 10.9) to 16.3 percentage points (95% CI, 9.9 to 22.6). Claude Haiku 4.5 reports lowered their accuracy, with paired differences ranging from −25.9 to −0.4 percentage points. Reading a reasoner's own reports improved accuracy with a CI excluding zero only for GPT-5 mini on SLAKE: 4.4 percentage points (95% CI, 0.9 to 8.2). Gemini 3.7 Flash had a paired difference of −9.2 percentage points (95% CI, −13.3 to −5.2) with its own reports on MedXpertQA-MM.

The reasoner changed accuracy even with identical reports on MedXpertQA-MM. Gemini 3.7 Flash achieved 60.8%, 74.5%, and 57.8% with GPT-5 mini, Gemini 3.7 Flash, and Claude Haiku 4.5 reports, respectively. Claude Haiku 4.5 achieved 30.6%, 45.6%, and 27.4% with the same

reports; GPT-5 mini achieved 50.1%, 62.0%, and 45.0%. Image-reading accuracy was 83.7%, 31.7%, and 52.9% for Gemini 3.7 Flash, Claude Haiku 4.5, and GPT-5 mini, respectively.

One shared report produced higher error correlation than separate reports from the same model. Across data sets and reasoners, error correlation ranged from 0.800 to 0.990 with one shared report, from 0.641 to 0.818 with own reports, and from 0.692 to 0.915 with images (Fig. 6b).

Reports omitted the queried SLAKE target among both errors and all reports. Omission affected 25.8% to 47.1% of errors and 26.5% to 30.8% of all reports. With Gemini 3.7 Flash as perceiver, these shares were 33.6% to 47.1% and 26.7%, respectively.

3.5. What two readers share

Readers sharing everything retained model-dependent error correlation. Across data sets, baseline correlations ranged from 0.692 to 0.800 for GPT-5 mini, 0.776 to 0.828 for Gemini 3.7 Flash, 0.878 to 0.915 for Claude Haiku 4.5, and 0.442 to 0.629 for Qwen 3.8 Flash (Fig. 7a).

Changing shared components lowered mean error correlation in the matched contrasts (Fig. 7b). In VQA-RAD, SLAKE, and MedXpertQA-MM order, role contrasts averaged −0.048 (95% CI, −0.071 to −0.023), −0.060 (95% CI, −0.087 to −0.036), and −0.033 (95% CI, −0.051 to −0.014). Rendering contrasts were −0.092 (95% CI, −0.123 to −0.068), −0.094 (95% CI, −0.123 to −0.066), and −0.045 (95% CI, −0.062 to −0.027). Reference-case contrasts were −0.110 (95% CI, −0.142 to −0.084) and −0.146 (95% CI, −0.179 to −0.117) on VQA-RAD and SLAKE. Report contrasts were −0.199 (95% CI, −0.233 to −0.164), −0.204 (95% CI, −0.247 to −0.162), and −0.172 (95% CI, −0.211 to −0.142). Perceiver contrasts were −0.311 (95% CI, −0.362 to −0.260), −0.275 (95% CI, −0.334 to −0.218), and −0.263 (95% CI, −0.294 to −0.227). Reasoner contrasts were −0.108 (95% CI, −0.132 to −0.083), −0.167 (95% CI, −0.222 to −0.129), and −0.430 (95% CI, −0.477 to −0.389). Model contrasts were −0.470 (95% CI, −0.522 to −0.421), −0.473 (95% CI, −0.544 to −0.406), and −0.492 (95% CI, −0.536 to −0.445).

Model-specific contrasts varied in direction for roles and rendering. VQA-RAD ranges were −0.148 to 0.011 and −0.181 to 0.023, respectively. Reference-case contrasts ranged from −0.229 to −0.043 on VQA-RAD and from −0.247 to −0.082 on SLAKE. MedXpertQA-MM reasoner contrasts ranged from −0.546 to −0.310; VQA-RAD model contrasts ranged from

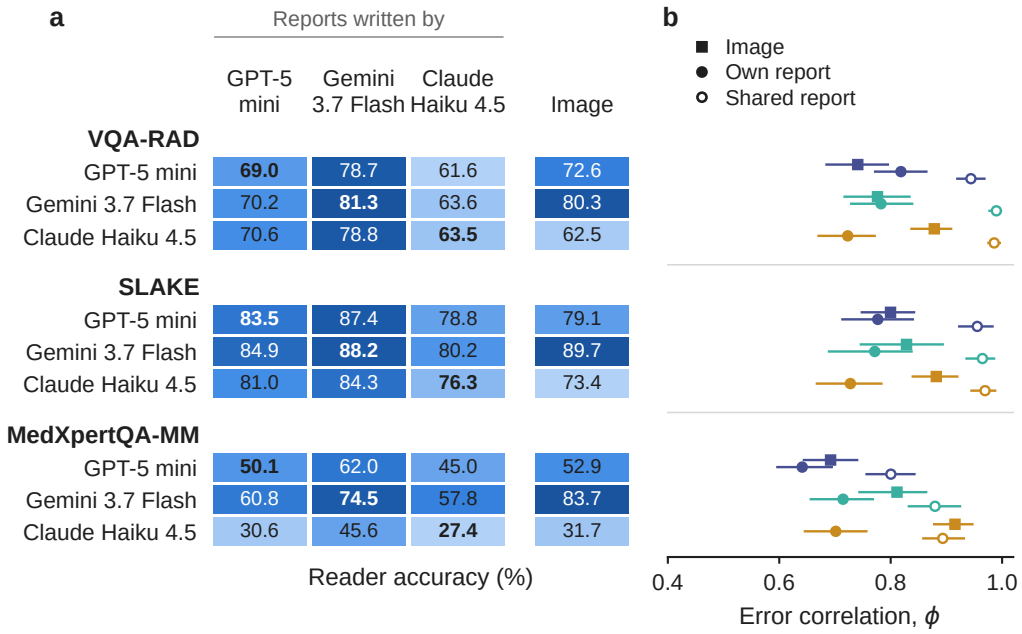


Figure 6: Reader accuracy and error correlation with question-blind findings reports. (a) Mean reader accuracy when each reader reads its own report from the column's perceiver without the image or reads the image directly. Rows identify reasoners within VQA-RAD, SLAKE, and MedXpertQA-MM (250 items each). Cell numbers give accuracy percentages; darker blue denotes higher accuracy within each data set, and bold numbers identify matching perceiver and reasoner models. (b) Error correlation, ϕ, among 9 readers. Filled squares denote image reading; filled circles, each reader's own report from the same model; open circles, one shared report from that model. Blue denotes GPT-5 mini; teal, Gemini 3.7 Flash; amber, Claude Haiku 4.5. Horizontal intervals are item bootstrap 95% CIs. Light gray horizontal lines separate data sets; no reference lines are plotted.

−0.696 to −0.272. The regression summary also associated different perceivers and reasoners with lower error correlation across data sets (Supplementary Tables).

3.6. Reference cases

Shared reference cases changed single-reader accuracy differently across models and data sets. On VQA-RAD, Claude Haiku 4.5 accuracy was 60.8% without reference cases, 68.8% with reference images, and 61.2% with reference text (Fig. 8a). On SLAKE, its corresponding accuracies were 72.4%, 67.6%, and 76.8%. GPT-5 mini and Gemini 3.7 Flash achieved 73.6% and

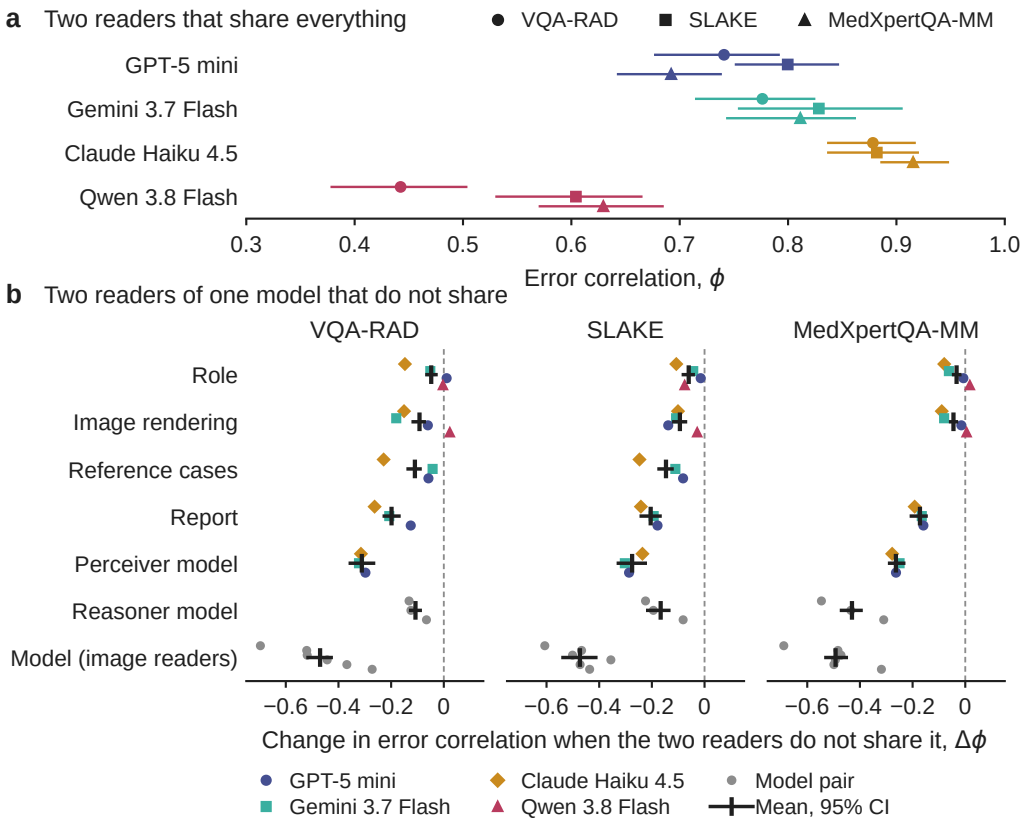


Figure 7: Error correlation by shared reader components. (a) Baseline error correlation, ϕ, between generic-panel readers sharing the model, image, rendering, role, and reference cases. Circles denote VQA-RAD; squares, SLAKE; triangles, MedXpertQA-MM (250 items each). Colors identify models. (b) Matched differences in mean pairwise error correlation, $\Delta\phi$, for reader pairs differing in the named component relative to pairs sharing it. Columns identify data sets; reference-case contrasts use VQA-RAD and SLAKE. Role compares specialty with generic panels; rendering compares different renderings with original images; reference cases compare private with shared cases; report compares own with shared reports. Perceiver compares reports from different perceivers with separate reports from one perceiver. Reasoner compares different reasoner models with one reasoner model reading a shared report. Model compares different models with one model reading images. Colored markers denote individual models: blue circles, GPT-5 mini; teal squares, Gemini 3.7 Flash; amber diamonds, Claude Haiku 4.5; rose triangles, Qwen 3.8 Flash. Gray circles denote model pairs for reasoner and model contrasts. Black vertical ticks denote means over models or model pairs. Horizontal intervals in (a) and black intervals in (b) are item bootstrap 95% CIs. Dashed vertical reference lines in (b) mark zero change.

82.4% with reference images on VQA-RAD, compared with 75.6% and 83.6% with reference text. Corresponding SLAKE accuracies were 79.2% and 89.2% with images, and 81.6% and 90.8% with text.

Private reference cases lowered error correlation relative to shared reference cases in every model and reference format, while panel accuracy varied (Fig. 8b,c). On SLAKE, GPT-5 mini panel accuracy was 81.6% with shared reference images and 84.0% with private reference images; error correlation was 0.741 and 0.667, respectively. Gemini 3.7 Flash panel accuracy was 90.4% with shared reference text and 92.0% with private reference text; error correlation was 0.823 and 0.716.

Claude Haiku 4.5 private reference cases lowered error correlation without raising panel accuracy in either format or data set. With SLAKE reference images, error correlation was 0.928 with shared reference cases and 0.655 with private reference cases, while panel accuracy was 70.4% and 68.8%. With reference text, error correlation was 0.941 and 0.720, while panel accuracy remained 78.0%.

3.7. Discussion among readers

On MedXpertQA-MM, readers abandoned correct answers that differed from their gray-image answer more often than correct answers that matched it. Abandonment ranged from 7.2% to 23.6% for the former and from 2.4% to 6.3% for the latter (Fig. 9a). Paired differences ranged from 4.6 percentage points (95% CI, 0.8 to 9.0) for Gemini 3.7 Flash to 17.4 percentage points (95% CI, 6.4 to 29.7) for Claude Haiku 4.5. All paired difference CIs excluded zero.

The difference in abandonment was smaller on VQA-RAD and SLAKE. GPT-5 mini had difference CIs excluding zero: 5.1 percentage points (95% CI, 0.4 to 10.4) and 2.6 percentage points (95% CI, 0.3 to 5.2), respectively (Fig. 9a).

Readers abandoned correct minority answers more often than correct majority answers. Across models and data sets, abandonment ranged from 41.5% to 100.0% for answers initially held by a minority and from 1.0% to 6.5% for majority answers (Fig. 9b).

Correct answers differing from the gray-image answer were more often in the initial minority on MedXpertQA-MM than on the other data sets. The share ranged from 6.6% to 23.5%, compared with 2.5% to 11.2% on VQA-RAD and 1.8% to 4.0% on SLAKE (Fig. 9c).

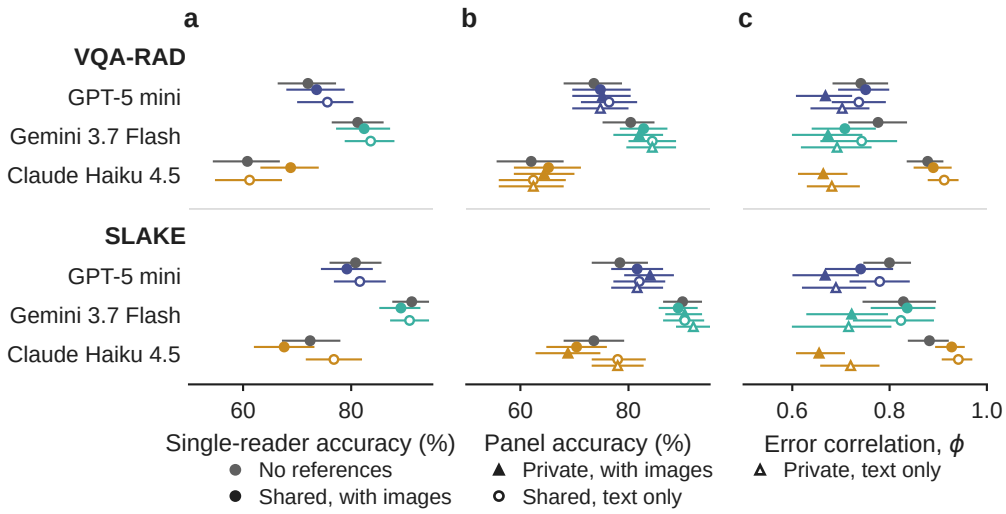


Figure 8: Accuracy and error correlation with reference cases. (a) Single-reader accuracy without reference cases, with shared reference images and question–answer pairs, or with the same pairs as text alone. (b) Majority-vote accuracy of generic panels of 9 readers. (c) Panel error correlation, ϕ. Rows group VQA-RAD and SLAKE (250 items each) by model. Gray filled circles denote no reference cases. Colored circles denote shared reference cases; triangles, private reference cases assigned separately to each reader. Filled colored markers denote reference cases with images; open markers, text-only reference cases. Blue denotes GPT-5 mini; teal, Gemini 3.7 Flash; amber, Claude Haiku 4.5. Horizontal intervals are item bootstrap 95% CIs. Light gray horizontal lines separate data sets; no reference lines are plotted.

3.8. Open-weight models

Open-weight models lost accuracy without the image on VQA-RAD and SLAKE, while MedXpertQA-MM accuracy stayed near chance. Original-image accuracy ranged from 56.0% to 80.0% on VQA-RAD and from 66.0% to 88.8% on SLAKE (Supplementary Fig. S1a). Gray-image ranges were 54.0% to 60.8% and 40.0% to 50.8%, respectively. On MedXpertQA-MM, original-image accuracy ranged from 20.0% to 24.4%, compared with 20.0% to 23.6% with gray images. MedGemma 4B had the highest original-image accuracy on VQA-RAD and SLAKE; Qwen3-VL 8B had the highest on MedXpertQA-MM.

Visual reliance varied across open-weight models. Mean absolute change in log-odds of yes ranged from 0.89 to 6.43 on VQA-RAD and from 1.40 to 11.23 on SLAKE (Supplementary Fig. S1b). On MedXpertQA-MM, mean

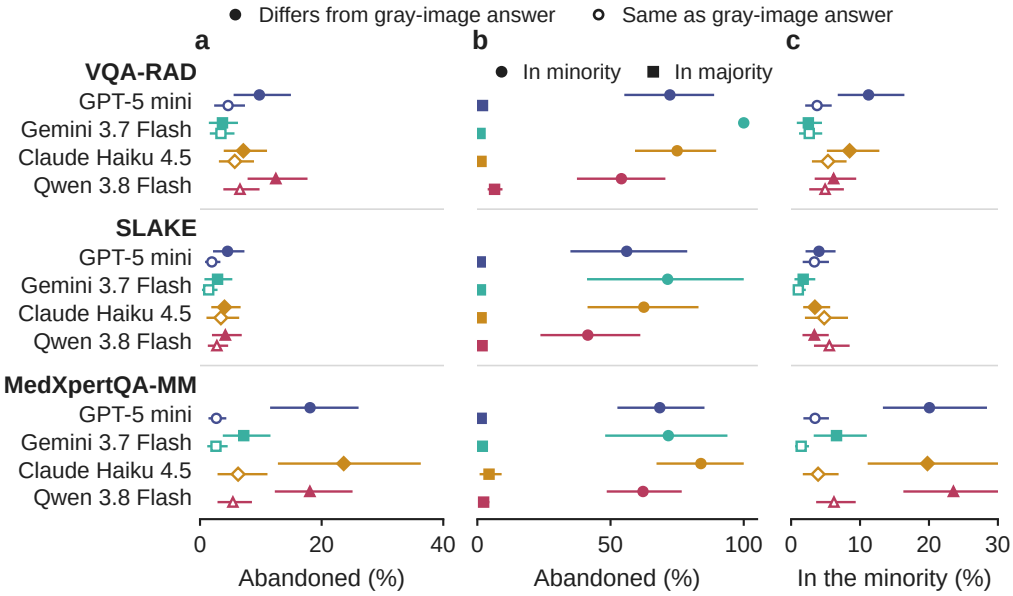


Figure 9: Correct answers abandoned during discussion. (a) Proportion of correct round-0 answers that become wrong, separated by whether the answer differs from or matches the same reader's gray-image answer. (b) Abandonment among correct round-0 minority and majority answers. (c) Proportion of correct answers initially held by a minority. Discussion panels contain 9 readers. Rows group VQA-RAD, SLAKE, and MedXpertQA-MM (250 items each). Filled and open markers in (a) and (c) denote answers differing from and matching the gray-image answer, respectively. Circles and squares in (b) denote minority and majority answers. Blue denotes GPT-5 mini; teal, Gemini 3.7 Flash; amber, Claude Haiku 4.5; rose, Qwen 3.8 Flash. In (a) and (c), model markers are circles, squares, diamonds, and triangles, respectively. No reference lines are plotted. Estimates are ratios of reader counts; horizontal intervals are item bootstrap 95% CIs.

visual reliance measured by Kullback–Leibler divergence ranged from 0.10 to 0.88.

Models sharing a vision encoder did not have consistently higher error correlation than models with different encoders. For HuatuoGPT-Vision 7B and LLaVA-1.5 7B, error correlation divided by its maximum given marginal error rates ranged from 0.213 to 0.451 (Supplementary Fig. S1c). Values for pairs with different encoders ranged from 0.148 to 0.725.

API consensus errors were associated with lower open-weight visual reliance on SLAKE, with little association on the other data sets. Spearman correlations were −0.048 on VQA-RAD, −0.401 on SLAKE, and 0.006 on MedXpertQA-MM.

4. Discussion

A reading panel of nine readers built from one VLM did not provide nine independent reads. Shared errors limited the benefit of majority voting, and discussion increased agreement without consistently improving accuracy. Differences in the findings available to readers and in the models processing them reduced error correlation more than changes in roles. The same error correlation and rise during discussion occurred for text-only clinical questions, from 0.758 to 0.990 (Supplementary S5). Shared errors therefore extend beyond image interpretation to the way readers built from the same model answer clinical questions.

The panel advantage depended on the accuracy of its readers and their error correlation. Qwen 3.8 Flash had less correlated errors than the other models and gained more over the single reader. On VQA-RAD, its mean panel reader exceeded the single reader by 10.3 percentage points, and aggregation added 3.3 percentage points. Across the yes/no panels, observed majority accuracy minus the beta-binomial prediction averaged -0.07 percentage points (Fig. 2). This agreement connects the limited return from additional readers to measured error correlation. Counting responses as independent observations overstates the information in a panel when the same items elicit errors from most readers.

Discussion preserved correct majority answers more often than correct minority answers. On MedXpertQA-MM, every model abandoned correct answers that differed from its gray-image answers more often than correct answers that matched them (Fig. 9). Correct answers dependent on image evidence were also more often in the initial minority on that set. Discussion thus exposed those answers to a majority sharing an incorrect question-text prior. After discussion, 8.9% to 70.0% of unanimous answers were wrong. Radiologists' deference to incorrect AI BI-RADS suggestions provides a clinical example of accepting another reader's assessment despite its error.[25] In these panels, the final consensus reflects both the initial readings and subsequent deference to peers.

The gray-image experiments connected shared errors to the question-text prior. Removing the image lowered accuracy toward chance on VQA-RAD and SLAKE and increased error correlation for Gemini 3.7 Flash, Claude Haiku 4.5, and Qwen 3.8 Flash (Fig. 3). On MedXpertQA-MM, most consensus errors repeated the gray-image majority, ranging from 64.2% to 76.2% across models (Fig. 4). Readers therefore often reached the same wrong an-

swer with and without image evidence. Language priors in visual question answering and poor performance under medical probing provide related evidence.[10,11] Clinical information also changes diagnostic test interpretation.[26] Here, question text supplied information that steered readers toward an answer even when the image required a different one.

Image interventions distinguished use of the relevant anatomy from answering correctly through the question text. Gemini 3.7 Flash had the greatest panel-reader region dependence on SLAKE-target, at 32.9 percentage points, compared with 3.2 percentage points for Claude Haiku 4.5 (Fig. 5). Target occlusion changed answers more than matched control occlusion, directly connecting responses to the queried region. The paired-image experiment held the question constant and required opposite answers for different images. Readers frequently gave the same answer to both. Together, these interventions explain why accuracy on isolated questions is insufficient evidence of image use, consistent with correct medical answers accompanied by flawed image comprehension.[12]

Seeing and judging each contributed to report-pathway accuracy (Fig. 6). Gemini 3.7 Flash reports improved GPT-5 mini and Claude Haiku 4.5 on every data set relative to reading images directly. Gemini 3.7 Flash as reasoner performed worse with its own reports on MedXpertQA-MM than with images. Given identical Gemini 3.7 Flash reports on that set, Gemini 3.7 Flash achieved 74.5% accuracy, GPT-5 mini achieved 62.0%, and Claude Haiku 4.5 achieved 45.6%. The full comparison therefore identifies both the perceiver's extraction of findings and the reasoner's selection of an answer as sources of performance differences. Introducing an intermediate report did not itself improve reading; the findings supplied and the model interpreting them determined the result.

Sharing a report made readers' errors more alike. Across reasoners and data sets, shared-report error correlation ranged from 0.800 to 0.990 and exceeded correlation with separate reports from the same model. Fixing the findings removed much of the variation between readers, locating an important source of their differences in seeing. Reports also omitted queried SLAKE targets among both correct and incorrect answers. A question-blind findings report makes the extracted findings available for inspection, while its omissions determine which image information reaches judging. Repeated judgments from a shared report consequently depend on the same account of the image.

The matched contrasts formed a graded ladder of reductions in error cor-

relation (Fig. 7). On VQA-RAD, mean changes over models were −0.048 for role, −0.092 for rendering, −0.110 for reference cases, −0.199 for report, −0.311 for perceiver, and −0.470 for model. Roles changed instructions; renderings changed the presentation of the same image; private reference cases changed the examples available to each reader. These interventions progressively changed more of the information used in a reading while retaining the underlying model. Separate reports supplied different findings, different perceivers changed the models extracting those findings, and different models changed both seeing and judging. The ordering also held on SLAKE and, for the evaluated contrasts, MedXpertQA-MM. None of the within-model changes brought error correlation near the level of readers from different models. Changing reasoners while holding the report fixed also reduced error correlation, with the greatest reduction on MedXpertQA-MM.

Complementary errors and adequate reader accuracy together determine the value of a second read. Private reference cases lowered error correlation across the evaluated models and formats, while panel accuracy varied (Fig. 8). Claude Haiku 4.5's SLAKE panel accuracy with text reference cases remained 78.0% under both shared and private assignment. Gemini 3.7 Flash's private text reference cases increased panel accuracy on SLAKE. Human evidence likewise supports pooling independent judgments of similar accuracy,[4] and double reading increased mammographic cancer detection.[1] For AI reading panels, the evidence supports evaluating a second reader by the errors it corrects and introduces, alongside its individual accuracy and correlation with the first reader. In radiology, a reading panel built from one model is closer to one read than to nine, and a second AI read adds independent information when a different model performs the seeing.

This study used closed questions from public data sets and included no human readers or clinical reporting. Each condition used a single sampling seed; item bootstrap intervals did not measure variation across seeds. API models were evaluated through a single router at one time point. Gray-image abstentions counted as wrong; correlation among answered items was included as a sensitivity analysis.

5. Conclusions

Reading panels built from the same VLM shared errors that limited the benefit of majority voting. Panel accuracy depended on both individual reader accuracy and error correlation. Question-text priors contributed to

shared errors, and a shared findings report increased error correlation. Different models reduced error correlation more than changes in roles, image rendering, or reference cases within a model. Discussion increased agreement and overturned correct minority answers, including answers dependent on image evidence. A reading panel of one model's readers is closer to one read than to nine, and a second AI read adds independent information when a different model performs the seeing.

Authors' contribution

TBD.

Ethics statement

The study used only public, de-identified data sets and involved no human participants. Institutional review board approval was not required.

Conflicts of interest

The authors declare no conflicts of interest.

References

[1] P. Taylor, H. W. W. Potts, Computer aids and human second reading as interventions in screening mammography: two systematic reviews to compare effects on cancer detection and recall rate, European Journal of Cancer 44 (6) (2008) 798–807. doi:10.1016/j.ejca.2008.02.016.

[2] H. Geijer, M. Geijer, Added value of double reading in diagnostic radiology, a systematic review, Insights into Imaging 9 (3) (2018) 287–301. doi:10.1007/s13244-018-0599-0.

[3] M. A. Bruno, E. A. Walker, H. H. Abujudeh, Understanding and confronting our mistakes: the epidemiology of error in radiology and strategies for error reduction, RadioGraphics 35 (6) (2015) 1668–1676. doi:10.1148/rg.2015150023.

[4] R. H. J. M. Kurvers, S. M. Herzog, R. Hertwig, J. Krause, P. A. Carney, A. Bogart, et al., Boosting medical diagnostics by pooling independent judgments, Proceedings of the National Academy of Sciences of the United States of America 113 (31) (2016) 8777–8782. doi:10.1073/pnas.1601827113.

[5] S. M. McKinney, M. Sieniek, V. Godbole, J. Godwin, N. Antropova, H. Ashrafian, et al., International evaluation of an AI system for breast cancer screening, Nature 577 (7788) (2020) 89–94. doi:10.1038/s41586-019-1799-6.

[6] Y. Kim, C. Park, H. Jeong, Y. S. Chan, X. Xu, D. McDuff, H. Lee, M. Ghassemi, C. Breazeal, H. W. Park, Mdagents: An adaptive collaboration of llms for medical decision-making, in: NeurIPS, 2024.

[7] X. Tang, A. Zou, Z. Zhang, Z. Li, Y. Zhao, X. Zhang, A. Cohan, M. Gerstein, Medagents: Large language models as collaborators for zero-shot medical reasoning, in: Findings of ACL, 2024.

[8] Y. Kim, K. Gu, C. Park, C. Park, S. Schmidgall, A. A. Heydari, Y. Yan, Z. Zhang, Y. Zhuang, Y. Liu, M. Malhotra, P. P. Liang, H. W. Park, Y. Yang, X. Xu, Y. Du, S. Patel, T. Althoff, D. McDuff, X. Liu, Capable language models can outgrow the benefits of collaboration, Nature Machine Intelligence 8 (2026) 1157–1172, preprint: arXiv:2512.08296, *Towards a Science of Scaling Agent Systems*. doi:10.1038/s42256-026-01268-y.

[9] E. Kim, A. Garg, K. Peng, N. Garg, Correlated errors in large language models, in: ICML, 2025.

[10] Y. Goyal, T. Khot, D. Summers-Stay, D. Batra, D. Parikh, Making the V in VQA matter: Elevating the role of image understanding in visual question answering, in: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2017, pp. 6904–6913.

[11] Q. Yan, X. He, X. Yue, X. E. Wang, Worse than random? An embarrassingly simple probing evaluation of large multimodal models in medical VQA, in: Findings of the Association for Computational Linguistics: ACL 2025, 2025.

[12] Q. Jin, F. Chen, Y. Zhou, Z. Xu, J. M. Cheung, R. Chen, et al., Hidden flaws behind expert-level accuracy of multimodal GPT-4 vision in medicine, npj Digital Medicine 7 (2024) 190. doi:10.1038/s41746-024-01185-7.

[13] A. Zafar, L. K. Murali, S. Bharadwaj, A. Vashist, J. Wu, Do medical vision language models actually see? A counterfactual grounding framework and hard-negative contrastive training for visually-reliant medical VLMs, arXiv:2607.03647 (2026).

[14] L. Butsanets, C. Corbière, J. Khlaut, P. Manceron, C. Dancette, RadImageNet-VQA: A large-scale CT and MRI dataset for radiologic visual question answering, in: Proceedings of the 9th International Conference on Medical Imaging with Deep Learning, Vol. 315 of Proceedings of Machine Learning Research, 2026, pp. 3036–3068.

[15] J. J. Lau, S. Gayen, A. Ben Abacha, D. Demner-Fushman, A dataset of clinically generated visual questions and answers about radiology images, Scientific Data 5 (2018) 180251. doi:10.1038/sdata.2018.251.

[16] B. Liu, L.-M. Zhan, L. Xu, L. Ma, Y. Yang, X.-M. Wu, SLAKE: A semantically-labeled knowledge-enhanced dataset for medical visual question answering, in: 2021 IEEE 18th International Symposium on Biomedical Imaging (ISBI), 2021.
URL https://arxiv.org/abs/2102.09542

[17] Y. Zuo, S. Qu, Y. Li, Z. Chen, X. Zhu, E. Hua, K. Zhang, N. Ding, B. Zhou, Medxpertqa: Benchmarking expert-level medical reasoning and understanding, in: ICML, 2025.

[18] A. Sellergren, S. Kazemzadeh, T. Jaroensri, A. Kiraly, M. Traverse, T. Kohlberger, et al., MedGemma technical report, arXiv preprint arXiv:2507.05201 (2025). doi:10.48550/arXiv.2507.05201.

[19] S. Bai, Y. Cai, R. Chen, K. Chen, X. Chen, Z. Cheng, et al., Qwen3-VL technical report, arXiv preprint arXiv:2511.21631 (2025). doi:10.48550/arXiv.2511.21631.

[20] P. Wang, S. Bai, S. Tan, S. Wang, Z. Fan, J. Bai, et al., Qwen2-VL: Enhancing vision-language model's perception of the world at any res-

olution, arXiv preprint arXiv:2409.12191 (2024). doi:10.48550/arXiv.2409.12191.

[21] J. Chen, C. Gui, R. Ouyang, A. Gao, S. Chen, G. H. Chen, et al., HuatuoGPT-Vision, towards injecting medical visual knowledge into multimodal LLMs at scale, arXiv preprint arXiv:2406.19280 (2024). doi:10.48550/arXiv.2406.19280.

[22] H. Liu, C. Li, Y. Li, Y. J. Lee, Improved baselines with visual instruction tuning, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2024.
URL https://arxiv.org/abs/2310.03744

[23] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, et al., Learning transferable visual models from natural language supervision, in: Proceedings of the 38th International Conference on Machine Learning, Vol. 139 of Proceedings of Machine Learning Research, 2021, pp. 8748–8763.
URL https://proceedings.mlr.press/v139/radford21a.html

[24] K. Zuiderveld, Contrast limited adaptive histogram equalization, in: P. S. Heckbert (Ed.), Graphics Gems IV, Academic Press, 1994, pp. 474–485. doi:10.1016/B978-0-12-336156-1.50061-6.

[25] T. Dratsch, X. Chen, M. Rezazade Mehrizi, R. Kloeckner, A. Mähringer-Kunz, M. Püsken, et al., Automation bias in mammography: the impact of artificial intelligence BI-RADS suggestions on reader performance, Radiology 307 (4) (2023) e222176. doi:10.1148/radiol.222176.

[26] C. T. Loy, L. Irwig, Accuracy of diagnostic tests read with and without clinical information: a systematic review, JAMA 292 (13) (2004) 1602–1609. doi:10.1001/jama.292.13.1602.